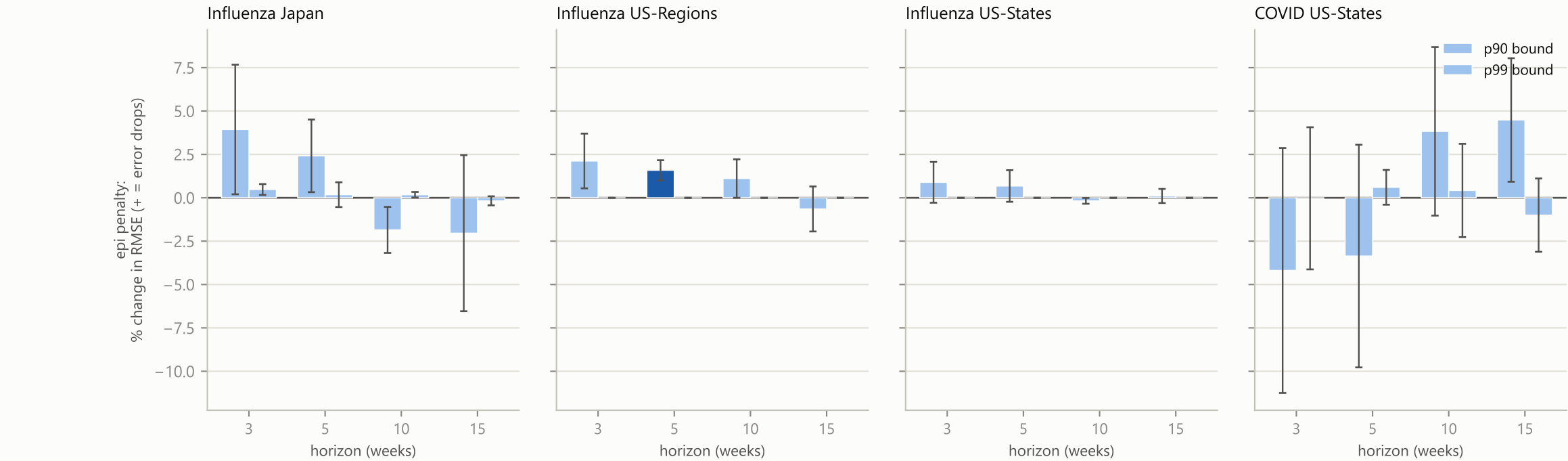


F3 Epidemiology-informed penalty. RMSE, lower is better; bars show the change from adding the penalty.



Paired by seed against the plain single-disease model, 5 seeds, same within-noise rule as the committed epi reports (a bound whose bar is pale is within noise). Two strictness bounds shown. Dengue was NOT run for this ablation and is absent by design, not omitted. Almost every cell sits within noise, so steering the model with a peak-intensity penalty neither helps nor hurts error.